

# The Methodology

Why most backtests lie, and the discipline that keeps yours honest.

## The core problem

Most trading backtests are wrong in the same direction: they look better than reality. Two forces cause it. First, **look-ahead bias** — the code uses information that wasn't available at the moment of the decision. Second, **design bias** — the human chose the universe, parameters, and rules already knowing how history turned out. The first is a bug you can fix; the second is a trap you can only disclose. This toolkit is built around taking both seriously.

## The five phases

The 25 skills map to the lifecycle of a serious quant research process. Each phase exists to catch a specific way strategies fool their builders.

| Phase                           | What it protects against                                                |
|---------------------------------|-------------------------------------------------------------------------|
| Data Integrity & Foundations    | Future-data leakage, survivorship, restated fundamentals, feed mismatch |
| Signal & Strategy Construction  | Signals that look predictive only because of leakage or hindsight       |
| Risk & Position Management      | Sizing and stops that flatter returns but blow up live                  |
| Backtesting & Validation        | In-sample fantasy, unrealistic fills, returns you can't explain         |
| Research Discipline & Iteration | Curve-fitting, p-hacking, and forgetting what you already tried         |

## The honesty principle

This toolkit deliberately refuses to sell a number. There is no "this returns X%" anywhere in it, because such a number would be a hindsight artifact, not a forecast. Instead it sells *rigor*: the ability to know whether a result is real. The two flagship skills — **lookahead-audit** and **anti-overfit** — exist precisely to tell you the uncomfortable truth before the market does.

A clean audit is not a profit guarantee. It only means your backtest didn't cheat. That is a far more valuable thing to own than a pretty equity curve.

## How the pieces fit

- **Skills** (markdown) teach Claude each task — install them into `~/ .claude/skills/`.
- **Excel templates** turn three skills into living worksheets: the EV planner, the training log, the metrics dashboard.
- **This documentation** and the interactive browser orient you and anyone you share the work with.

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